On Counting Cards and Learning Optimal Deviations from Blackjack Strategies

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Concrete

We can play blackjack according to a basic strategy, which is well-known to be optimal. In this paper we present an optimaler strategy which is not yet well-known, and more generally, the method to create your own optimaler strategy. This might tie into existing work on card counting, and paves the way for our methodology to be applied to many areas, such as finance, engineering, philosophy, algebraic geometry, geometric algebra, geometric geometry, algebraic algebra, and behavioural ergonomics.

Contents

1 Introduction

Blackjack is a popular psychological thriller roguelike played in the casino against a persistent boss called the dealer. The objective of a rational player in a round

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is to obtain a larger number of points than the dealer by choosing from a number of actions, without exceeding 21 (in which case the dealer wins). This includes but is not limited to hitting (obtaining another card), standing (ending your turn), doubling (getting exactly one more card for doubling down on the bet, not doubling your points), splitting (splitting your two cards into two hands), flicking, spinning, pulling, twisting, and bopping. The dealer is an example of artificial intelligence (AI). The AI strategy implemented by the dealer is in particular deterministic and known to the player, usually hitting until they have 17 points, when they then stop. Since the player goes first, the dealer has a slight statistical edge, called the house edge, due to their lower chance of busting I think. Card counting is a feature of the game of blackjack, which can be exploited to predict future card drops, or at least their approximate point value, in order to regain this statistical edge. Card counting combined with good bankroll management and bet sizing, some players can make quite a lot of money from the casino to play even more blackjack. The majority of blackjack players however, even (sometimes especially) those who know the basics of card counting, are not profitable. Interestingly this doesn't stop many blackjack players from thinking that actually they are good, an effect which is observed in other games, such as Mario Kart Wii, despite the fact that we[1] are almost surely better than the reader at Mario Kart Wii.

1.1 How to Count Cards and Win Loads and Loads and Loads of Money

Card counting, unlike what the name suggests, does not count the cards that have been played, but instead assigns values to the cards and tracks the running total of those values over the course of the game. This quantity, called the "running count" (RC), is divided by the approximate number of decks remaining in the shoe, to get the "true count" (TC). On top of what is colloquially known as basic strategy—which is the strategy which minimises house edge without information of what remains in the deck[2]—it is the TC which allows players to consistently obtain high scores. A popular example of a card counting system is called Hi-Lo, where $+1$ is added to the RC whenever a 2, 3, 4, 5, or 6 is revealed from the shoe, or -1 if you see a 10, J, Q, K, or A. This is probably popular because it's really easy. Different card counting systems differ in what is added to the RC, and all are meant to be linked to gains and losses to the house edge after the card is removed from the deck, and inform players how much to bet, and sometimes deviate from the basic strategy.

[1] By "we" in this paper, we always mean "I", as we are the only author.

[2] Since there is probability involved and something is being minimised, blackjack's basic strategy is also in fact an example of AI.

1.2 World Record Progression[3]

In the 1950s, coinciding with when humans first discovered AI, people began to try to use probability and statistics to their advantage in blackjack. The same can be said for poker, an easier version of blackjack, which is historically a game about eye contact. This decade saw the development of simple RC-only strategies. These strategies, while simple, were a significant leap forward in regaining the edge from the house. In the next decade, we saw the publication of [Tho66], who told people how to beat the dealer with statistics. The conditions were very favourable for Thorp from a statistical point of view—playing with only a single deck of cards until they've all run out, for example—but it was actually legal for casinos to murder card counters in this time, all the way up until January 1st, 1970. It is conjectured that the principles Thorp published were already in circulation within certain groups. These techniques focused on tracking the proportion of high cards to low cards in the deck, allowing for the bettor to bet more when this proportion is sufficiently skewed. The Hi-Lo system, along with its very many variations, became the de facto standard. Sadly for the reader, but the complete and utter opposite of sadly for the casino, simple systems such as the Hi-Lo system are really easy to detect by surveillance with modern technology.

The decades following the surge of interest in "advantage play" that Thorp described saw a shift from disruptive changes (card counting in general) to meticulous and incremental improvements. The focus moved from the Hi-Lo system's relative simplicity to the pursuit of strategies that offered a larger edge. This was the era of the number crunchers. The development of more granular counting systems, such as the Zen count, or Omega II, were developed for this purpose, and there is not necessarily a right answer on which method is best in practice.

This period also saw team play emerge, as it was found that casinos were paying a lot of attention to catching counters. While at the time being incredibly secretive, these groups sought to maximize their advantage by combining the skills of multiple individuals. For example, members of the team would play at different tables keeping track of the count and betting small, then somehow call to a designated member to join their table when TC is very high, with therefore very favourable conditions to the new player, who would then bet in very large amounts. The MIT blackjack team is a famous example of this approach. The use of computer simulations became ubiquitous during this time too, which could calculate expected value of a strategy much faster than a human could, even some people who were at MIT.

Coming up to the present day, many blackjack solvers exist, and can now compute expected values faster than *any* MIT student. It is believed that there are hundreds of thousands, if not hundreds, of blackjack solvers created

[3] It is recommended that the reader listens to "We're Finally Landing" [Hom16] for this section.[4]

[4] In fact, as is shown by the success of some authors, any literature review could be made less boring by listening to this. See for example [Sal21b; Sal21a; Ast09].

by students to demonstrate to employers that they are seriously passionate about writing their own Python code. Don't quote me on this, but blackjack is probably completely solved by now, at least mathematically. There is still money to be made in card counting, and there are undoubtedly a handful of people making a decent living off of it, likely with their edge being in clever teamwork and staying under the radar in novel ways.

Future directions of card counting are comprehensively summarised in Section 4 where we discuss directions to extend our work.

2 Methodology

This paper develops a novel and groundbreaking algorithm to modify blackjack's basic strategy. A keen-eyed reader may remark that blackjack's basic strategy is optimal. While this is true, note that ours will be optimaler.

An example of a basic strategy chart is shown in Figure 1. Reading the chart tells you how you should act in order to maximise expected value. Unfortunately however, sometimes following this chart does not result in winning the hand. This highlights a fundamental issue that I doubt has been addressed by the existing literature: players want to win *money*. Rent can't be paid in expected value. Could you imagine that? "Hey landlord, I don't have any money, but I have an edge that means odds are in my favour." The landlord might murder you, or worse: evict you.

We define this in Algorithm 1, but we describe our method to modify the basic strategy briefly here. We begin playing a hand against the dealer. We read from the basic strategy chart how to act. If this results in us winning, then we leave the chart untouched. However, if we lose, then we put the cards back the way they were, and try a different action. If that results in us winning, then we will modify the chart to read the new action instead of the old action.

3 Results

We train our adjusted basic strategy on a popular online casino site. When not enough information is known after a hand in order to determine which is the best action that we could have taken, we use "vibes" to determine whether or not we should have won. We present our findings in Figure 4. Our results show that basically sometimes you should not follow the basic strategy, but actually you should sometimes follow a different one (ours).[5] Remarkably—but also of course expectedly, since our research is sound—during training our algorithm, we profited £30.

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[5] 

DEALER'S UP CARD

PLAYER'S HAND	2	3	4	5	6	7	8	9	10	A
17+	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND
16	STAND	STAND	STAND	STAND	STAND	HIT	HIT	HIT	HIT	HIT
15	STAND	STAND	STAND	STAND	STAND	HIT	HIT	HIT	HIT	HIT
14	STAND	STAND	STAND	STAND	STAND	HIT	HIT	HIT	HIT	HIT
13	STAND	STAND	STAND	STAND	STAND	HIT	HIT	HIT	HIT	HIT
12	HIT	HIT	STAND	STAND	STAND	HIT	HIT	HIT	HIT	HIT
11	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT
10	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT	HIT
9	HIT	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT	HIT
5-8	HIT	HIT	HIT	HIT	HIT	HIT	HIT	HIT	HIT	HIT
A 8-10	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND
A 7	STAND	DOUBLE	DOUBLE	DOUBLE	DOUBLE	STAND	STAND	HIT	HIT	HIT
A 6	HIT	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 5	HIT	HIT	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 4	HIT	HIT	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 3	HIT	HIT	HIT	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 2	HIT	HIT	HIT	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A A	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT
10 10	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND
9 9	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	STAND	SPLIT	SPLIT	STAND	STAND
8 8	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT
7 7	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT
6 6	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT	HIT
5 5	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT	HIT
4 4	HIT	HIT	HIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT	HIT
3 3	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT
2 2	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT

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Figure 1: This basic strategy is presented in [Tea23]. Looking up your hand (rows) against the face-up card of the dealer lets you read off what the best decision is that you can make, in terms of expected value.

4 Conclusion and Future Work

In this paper, we presented a sweet algorithm to make a boatload of cash, inspiring the next generation of card counters. Actually, we sort of forgot to do any card counting, so we'll quickly just mention that there were always four cards to begin with, often with a higher number of cards by the end.

There are many directions this research could be taken further. For example, the use of transformers, such as the AI ones, could be used to modify the basic strategy, which would be much more scalable than what we did. Or, more generally than transformers, one could use a deep neural network. See Appendix A for more on this, as well as some proofs nobody will read.

References

[Ast09] Rick Astley. "Never Gonna Give You Up (Official Music Video)". In: *YouTube* (2009). URL: https://www.youtube.com/watch?v=dQw4w9WgXcQ.

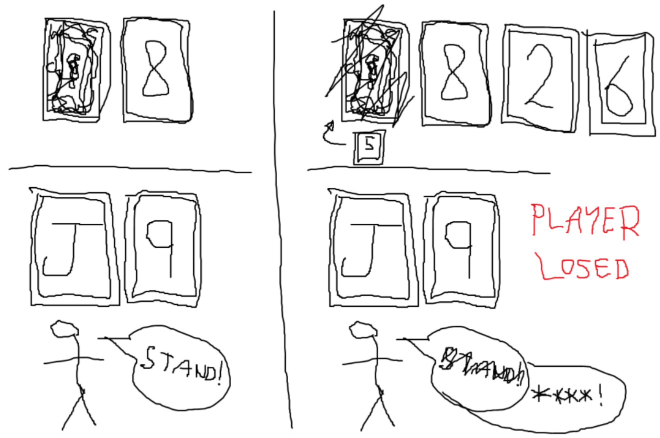

Figure 2: What happens when you follow the basic strategy chart and stand on 19 against dealer's 8.

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Figure 3: What happens when you deviate from the basic strategy chart and hit 19 against dealer's 8.

A DNNs as a Generalisation of Transformers

In this section, we prove that transformers are basically a special case of the humble fully connected feedforward deep neural network. Due to the universal approximation theorem I think, a sufficiently large DNN can approximate anything. Since transformers are something, using the fact that by definition, anything is the set of all somethings, we are done. For the remainder of this section, due to the fact that nobody reads these things, we have asked a very popular collaborator of many of our colleagues to generate the proofs of the theorems that we haven't stated here, which will now follow.

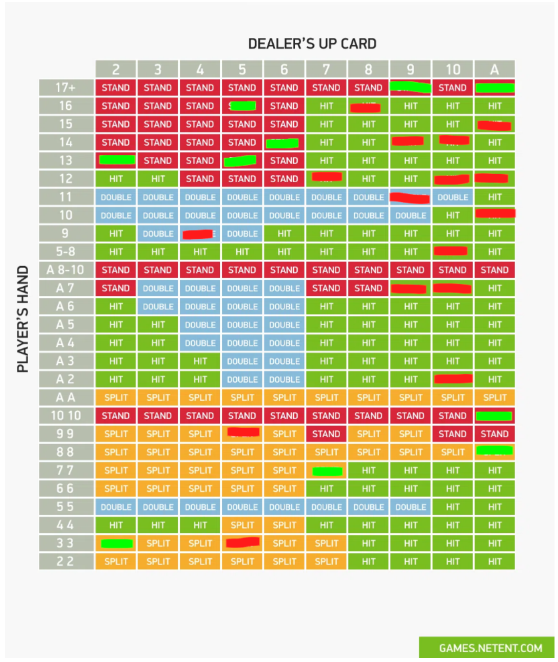
Let θ_t be the parameters at iteration t. We define the gradient as $g_t = \nabla\mathcal{L}(\theta_t)$. Assumption 1: $\mathcal{L}(\theta)$ is L-smooth, i.e., $\|\nabla\mathcal{L}(\theta_1) - \nabla\mathcal{L}(\theta_2)\| \leq L\|\theta_1 - \theta_2\|$ for all θ_1, θ_2. Assumption 2: The gradient is bounded, i.e., $\|g_t\| \leq G$ for all t. Assumption 3: The learning rate η_t satisfies $\sum_{t=1}^{\infty} \eta_t = \infty$ and $\sum_{t=1}^{\infty} \eta_t^2 < \infty$. Update rule: $\theta_{t+1} = \theta_t - \eta_t g_t$. Since $\mathcal{L}(\theta)$ is bounded below, a_{T+1} is bounded. Thus,

$$\sum_{t=1}^{T} \eta_t \|g_t\|^2 \left(1 - \frac{L\eta_t}{2}\right) < \infty.$$

Since $\sum_{t=1}^{\infty} \eta_t^2 < \infty$, for sufficiently large t, $1 - L\eta_t/2 > 1/2$. Thus,

$$\sum_{t=1}^{\infty} \eta_t \|g_t\|^2 < \infty.$$

By Assumption 3, $\sum_{t=1}^{\infty} \eta_t = \infty$. Therefore, $\lim_{t\to\infty} \|g_t\|^2 = 0$. Hence, $\lim_{t\to\infty} \|\nabla\mathcal{L}(\theta_t)\| = 0$. Let $X \in \mathbb{R}^{n\times d}$ be the input data matrix, and $Y \in \mathbb{R}^{n\times c}$ be the target ma-



DEALER'S UP CARD

PLAYER'S HAND

	2	3	4	5	6	7	8	9	10	A
17+	STAND	STAND	STAND	STAND	STAND	STAND	STAND		STAND	
16	STAND	STAND	STAND		STAND	HIT		HIT	HIT	HIT
15	STAND	STAND	STAND	STAND	STAND	HIT	HIT	HIT	HIT	
14	STAND	STAND	STAND	STAND		HIT	HIT			HIT
13		STAND	STAND		STAND	HIT	HIT	HIT	HIT	HIT
12	HIT	HIT	STAND	STAND	STAND		HIT	HIT		
11	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE		DOUBLE	HIT
10	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT	
9	HIT	DOUBLE		DOUBLE	HIT	HIT	HIT	HIT	HIT	HIT
5-8	HIT	HIT	HIT	HIT	HIT	HIT	HIT	HIT		HIT
A 8-10	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND
A 7	STAND	DOUBLE	DOUBLE	DOUBLE	DOUBLE	STAND	STAND			HIT
A 6	HIT	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 5	HIT	HIT	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 4	HIT	HIT	DOUBLE	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 3	HIT	HIT	HIT	DOUBLE	DOUBLE	HIT	HIT	HIT	HIT	HIT
A 2	HIT	HIT	HIT	DOUBLE	DOUBLE	HIT	HIT	HIT		HIT
A A	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT
10 10	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	STAND	
9 9	SPLIT	SPLIT	SPLIT		SPLIT	STAND	SPLIT	SPLIT	STAND	STAND
8 8	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	
7 7	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT		HIT	HIT	HIT	HIT
6 6	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT	HIT
5 5	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	DOUBLE	HIT	HIT
4 4	HIT	HIT	HIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT	HIT
3 3		SPLIT	SPLIT		SPLIT	SPLIT	HIT	HIT	HIT	HIT
2 2	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	SPLIT	HIT	HIT	HIT	HIT

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Figure 4: The result of our methodology on learning better strategies. The reader is encouraged to pretend that we put in all the text but you can definitely work out what we mean.

trix. We define the loss function as $\mathcal{L}(W) = \frac{1}{2n}\|XW - Y\|_F^2 + \frac{\lambda}{2}\|W\|_F^2$, where $W \in \mathbb{R}^{d\times c}$ is the weight matrix Taking the gradient with respect to W:

$$\begin{aligned}\nabla_W \mathcal{L}(W) &= \frac{1}{n}X^T(XW - Y) + \lambda W \\ &= \frac{1}{n}X^T XW - \frac{1}{n}X^T Y + \lambda W.\end{aligned}$$

Using the L-smoothness, we have:

$$\begin{aligned}\mathcal{L}(\theta_{t+1}) &\le \mathcal{L}(\theta_t) + \langle g_t, \theta_{t+1} - \theta_t\rangle + \frac{L}{2}\|\theta_{t+1} - \theta_t\|^2 \\ &= \mathcal{L}(\theta_t) - \eta_t\|g_t\|^2 + \frac{L}{2}\eta_t^2\|g_t\|^2 \\ &= \mathcal{L}(\theta_t) - \eta_t\|g_t\|^2\left(1 - \frac{L\eta_t}{2}\right).\end{aligned}$$

Let $a_t = \mathcal{L}(\theta_t)$. Then $a_{t+1} \le a_t - \eta_t\|g_t\|^2(1 - L\eta_t/2)$. Summing from $t = 1$ to

T:

$$\begin{aligned}\sum_{t=1}^{T}(a_{t+1}-a_t) &\leq -\sum_{t=1}^{T}\eta_t\|g_t\|^2\left(1-\frac{L\eta_t}{2}\right)\\ a_{T+1}-a_1 &\leq -\sum_{t=1}^{T}\eta_t\|g_t\|^2\left(1-\frac{L\eta_t}{2}\right).\end{aligned}$$

Setting the gradient to zero:

$$\begin{aligned}\frac{1}{n}X^TXW-\frac{1}{n}X^TY+\lambda W=0\\ \left(\frac{1}{n}X^TX+\lambda I\right)W=\frac{1}{n}X^TY\\ W=\left(\frac{1}{n}X^TX+\lambda I\right)^{-1}\frac{1}{n}X^TY.\end{aligned}$$

Let $A=\frac{1}{n}X^TX$. Then $W=(A+\lambda I)^{-1}\frac{1}{n}X^TY$. Using the Woodbury matrix identity, if $A=U\Sigma U^T$, then $(A+\lambda I)^{-1}=U(\Sigma+\lambda I)^{-1}U^T$. Thus, $W=U(\Sigma+\lambda I)^{-1}U^T\frac{1}{n}X^TY$. Now, let $Z=XW-Y$. We have $\|Z\|_F^2=\mathrm{tr}(Z^TZ)$.

$$\begin{aligned}\|Z\|_F^2 &= \mathrm{tr}((XW-Y)^T(XW-Y))\\ &= \mathrm{tr}(W^TX^TXW-2W^TX^TY+Y^TY).\end{aligned}$$

Substituting $W=(A+\lambda I)^{-1}\frac{1}{n}X^TY$, we get:

$$\begin{aligned}\|Z\|_F^2=\mathrm{tr}&\left(\left(\frac{1}{n}Y^TX(A+\lambda I)^{-1}\right)X^TX\left((A+\lambda I)^{-1}\frac{1}{n}X^TY\right)\right)\\ &-2\mathrm{tr}\left(\left(\frac{1}{n}Y^TX(A+\lambda I)^{-1}\right)X^TY\right)+\mathrm{tr}(Y^TY).\end{aligned}$$

Let $C=X^TYY^TX$.

$$\|Z\|_F^2=\frac{1}{n^2}\mathrm{tr}(X^TXBCB)-\frac{2}{n}\mathrm{tr}(CB)+\mathrm{tr}(Y^TY).$$

Let $B=(A+\lambda I)^{-1}$. Then,

$$\|Z\|_F^2=\mathrm{tr}\left(\frac{1}{n^2}Y^TXBX^TXBX^TY\right)-\frac{2}{n}\mathrm{tr}(Y^TXBX^TY)+\mathrm{tr}(Y^TY).$$

Applying the cyclic property of trace:

$$\|Z\|_F^2=\frac{1}{n^2}\mathrm{tr}(X^TXBX^TYY^TXB)-\frac{2}{n}\mathrm{tr}(X^TYY^TXB)+\mathrm{tr}(Y^TY).$$

Algorithm 1 [EE50]; see also [Eng50]

this
1 of 3
pronoun
plural: these
1a(1): the person, thing, or idea that is present or near in place, time, or thought or that has just been mentioned
these are my hands
1a(2): what is stated in the following phrase, clause, or discourse
I can only say *this*: it wasn't here yesterday
1b: this time or place
expected to return before *this*
2a: the one nearer or more immediately under observation or discussion
this is iron and that is tin
2b: the one more recently referred to

this
2 of 3
adjective
plural: these
1a: being the person, thing, or idea that is present or near in place, time, or thought or that has just been mentioned
this book is mine
early *this* morning
b: constituting the immediately following part of the present discourse
c: constituting the immediate past or future
friends all *these* years
d: being one not previously mentioned, used especially in narrative to give a sense of immediacy or vividness
then *this* guy runs in
had *this* urge to go shopping
2: being the nearer at hand or more immediately under observation or discussion this car or that one

this
3 of 3
adverb
plural: these
1: to the degree or extent indicated by something in the immediate context or situation
didn't expect to wait *this* long